

Homework 05

For this homework, create an **Quarto Markdown** (e.g., `.qmd`) file and save it to your `dataviz-exercises` project with the name and a prefix `hw05-lastname-firstInitial`. Full name example: `hw05cookg.qmd`. If you collaborate, add collaborator names as text in a header of your `.qmd` file.

Relevant Modules:

- data frame manipulation and wrangling
- the grammar of graphics
- coordinates, axes, and position scales
- statistical transformations

This homework is an exercise designed to challenge you to fix x and y axis scales of plots. The plot replication involves inspecting a plot, wrangling data and creating a data subset to prepare the data for plotting, selecting geoms, adding plot layers, and applying colors to specific data sub groups.

You will build a Quarto page with the answers/replication to the questions. I recommend practicing creating `.png` files with your plot-saving script and rendering them in your `.qmd` file. Doing so will ensure you know that your paths will allow for reproduction on a computer other than your own. However, but you may also use a code block in your `.qmd` file if you wish.

Used header markers to identify the three Problems. Render the page when completed.

Problems:

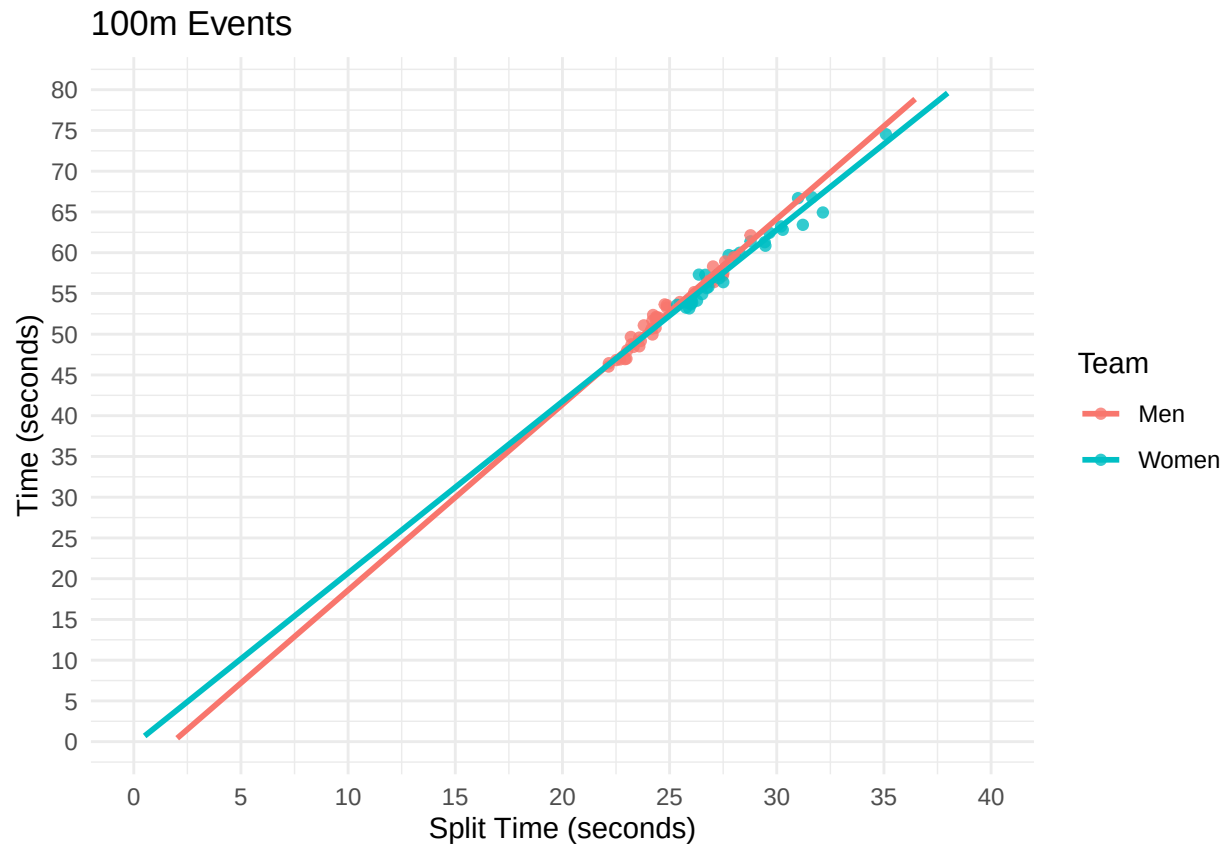
1. Replicating a Visualization

Getting the Data:

```
readr::read_csv(  
  "https://raw.githubusercontent.com/slicesofdata/dataviz25/main/data/processed/  
  cleaned-2023-cms-invite.csv")
```

The Plot

Feel free to collaborate. Consider the following description of the data to help. *For all events in the season, the event split times and completion times appear linearly related for both male and female swimmers.* Examine the data frame and study the plot to determine how you might go about replicating it and do so as best as you are able. Pay attention to issues related to recent topics, like scale limits and breaks.



2. Understanding Coordinate Modifications

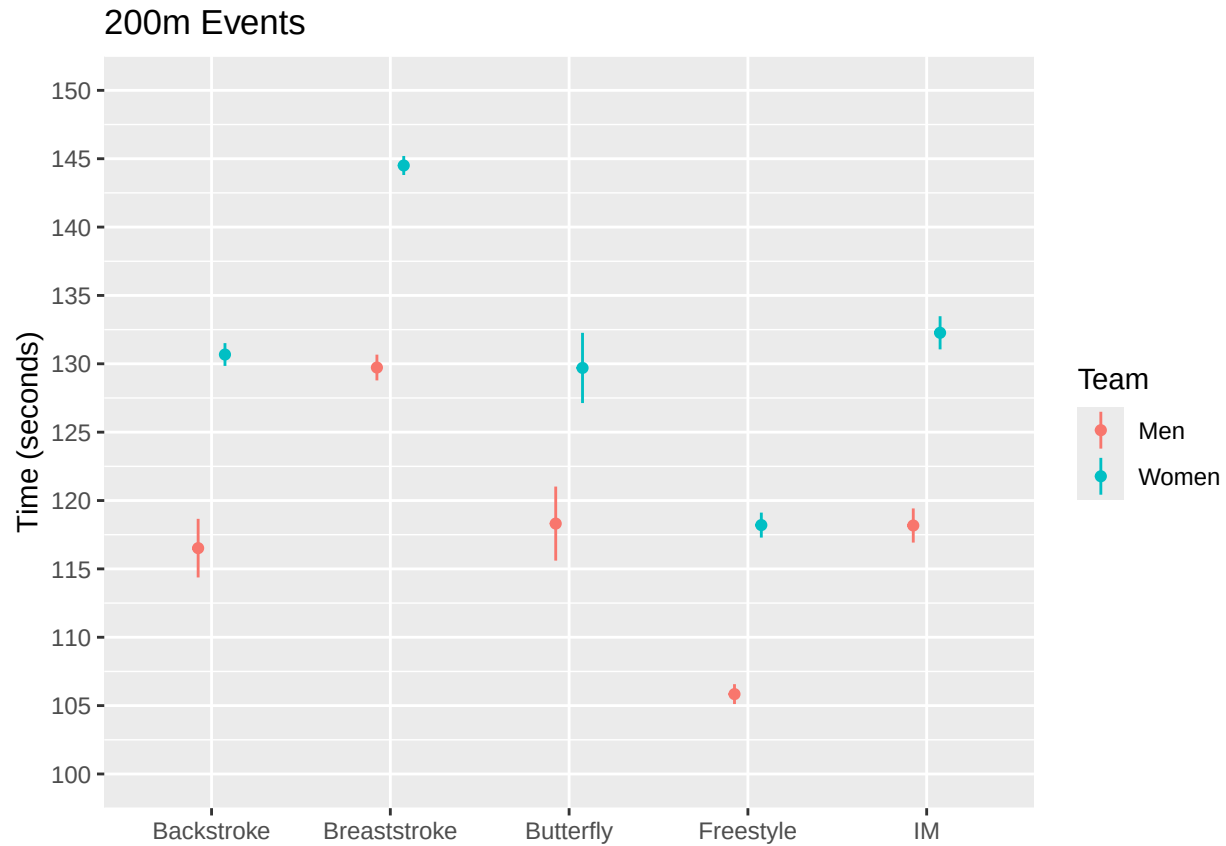
After replicating the plot, add a `coord_equal()` to make the unit change on y be equal to that on x.

Does this modification change how you perceive the data?

3. Replicating a Visualization

The Plot

Consider the following very general description of the data to help. *For all events in the season, the mean and standard deviations reveal more variability among butterfly event times for both men and women compared with other events.* Look at the data frame and study the plot to determine how you might go about replicating it. Replicate this plot as best as you are able.



Finally, render your `html` file and upload to: <https://ln5.sync.com/dl/a038628f0/wwfjxk-f7rfshin-rkedi3y8-77f9zaii>